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Casting Defect Detection: Robustness Analysis of HOG+LBP+SVM Pipeline Against Gaussian Noise with Autoencoder-Based Defect Localization

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Abstrak

Industri pengecoran logam sangat bergantung pada inspeksi kualitas untuk memastikan produk bebas dari cacat permukaan yang berpotensi menyebabkan kegagalan komponen. Penelitian ini mengkaji ketahanan sistem ekstraksi fitur manual yang menggabungkan deskriptor Histogram of Oriented Gradients dan Local Binary Pattern dengan pengklasifikasi Support Vector Machine. Sistem ini dievaluasi dalam menghadapi dua penurunan kualitas operasional utama, yakni derau sensor Gaussian dan distorsi spasial. Metode yang diusulkan beroperasi melalui dua tahap berurutan. Tahap pertama mengevaluasi lima skenario lintas domain dengan membandingkan dua model pengklasifikasi. Model pertama dilatih pada citra bersih, sementara model kedua dilatih pada citra berderau yang diproses melalui metode penyaringan Bilateral Filter dan Non Local Means. Tahap kedua menggunakan Convolutional Autoencoder yang dilatih secara tidak terawasi pada citra normal untuk melokalisasi area cacat melalui peta perbedaan rekonstruksi dan penentuan ambang batas persentil adaptif. Eksperimen dilakukan menggunakan dataset industri nyata yang berisi 7348 citra skala abu abu dari impeler pompa submersibel. Hasil tahap pertama menunjukkan model awal mencapai akurasi 99,16 persen pada data bersih, tetapi turun tajam menjadi 60,84 persen saat terpapar derau tanpa penyesuaian. Model yang diperkuat dengan strategi penyaringan yang tepat berhasil memulihkan akurasi hingga 98,60 persen. Pada kasus distorsi spasial, kedua model mencapai akurasi 76,00 persen dengan skor F1 di atas 80 persen. Tahap kedua sukses melokalisasi seluruh 200 sampel cacat yang diuji dengan rata rata 5,91 kotak pembatas per citra. Hal ini membuktikan besarnya potensi sistem untuk diterapkan pada lingkungan industri sesungguhnya.

Kata Kunci *HOG; LBP; SVM; noise Gaussian; denoising; Convolutional Autoencoder; lokalisasi anomali; ketahanan model*

Abstract

The metal casting industry relies heavily on quality inspection to ensure products are free from surface defects that could cause component failure. This research examines the robustness of a manual feature extraction pipeline combining Histogram of Oriented Gradients and Local Binary Pattern descriptors with a Support Vector Machine classifier. The system is evaluated against two main operational degradations which are Gaussian sensor noise and spatial distortion. The proposed method operates in two sequential stages. The first stage evaluates five cross domain scenarios by comparing two classification models. The initial model is trained on clean images, while the second uses noisy images processed through Bilateral Filter and Non Local Means denoising methods. The second stage utilizes a Convolutional Autoencoder trained unsupervised solely on normal images to localize defect areas through reconstruction difference maps and adaptive percentile thresholding. Experiments were conducted on a real industrial dataset containing 7348 grayscale images of submersible pump impellers. First stage results show the initial model achieved 99.16 percent accuracy on clean data but dropped drastically to 60.84 percent when exposed to noise without domain adaptation. The model reinforced with an appropriate denoising strategy successfully restored accuracy to 98.60 percent. Under spatial distortion, both models achieved 76.00 percent accuracy with an F1 score above 80 percent. The second stage successfully localized defects in all 200 evaluated defect samples with an average of 5.91 bounding boxes per image. These results demonstrate the strong potential of the system for implementation in actual industrial environments.

Keywords *HOG; LBP; SVM; Gaussian noise; denoising; Convolutional Autoencoder; anomaly localization; robustness evaluation*

1. Introduction

1.1 Background

The metal casting industry is a cornerstone of modern manufacturing, producing critical components - from submersible pump impellers to automotive parts - through processes that subject raw metal to extreme temperatures and pressures. Quality inspection is therefore indispensable: even microscopic surface defects such as blow holes, sand inclusions, or hairline cracks can cause component failure in service, resulting in costly recalls, safety risks, and reputational damage [1]. Traditionally, this inspection task has been performed manually by trained operators, an approach that is inherently subjective, slow, and difficult to scale across high-volume production lines [2].

Digital image processing offers a principled alternative. Conventional hand-crafted feature extraction methods - particularly Histogram of Oriented Gradients (HOG) and Local Binary Pattern (LBP) - have demonstrated strong effectiveness for defect detection in metallic surfaces [3]. HOG captures gradient orientation distributions that encode shape anomalies and boundary irregularities, while LBP encodes micro-surface texture patterns sensitive to porosity and surface roughness [4]. When fed into a Support Vector Machine (SVM) classifier with an RBF kernel, these combined features yield high classification accuracy on well-curated datasets [5]. However, a significant gap exists between laboratory performance and real-world deployment: factory-floor cameras, exposed to heat, vibration, and particulate contamination, progressively degrade image quality through the introduction of sensor noise [6]. Gaussian noise - the most pervasive form arising from sensor electronics - directly corrupts the gradient information on which HOG descriptors fundamentally depend [4], making noise robustness an essential but frequently overlooked requirement in production inspection systems.

In parallel, a second limitation of conventional classification-only systems is their inability to indicate where a defect is located within the image, providing only a binary verdict. For operators and engineers, spatial localization - knowing which region of the casting surface carries the anomaly - is actionable information that a single class label cannot convey. Unsupervised anomaly detection via Convolutional Autoencoders (CAE) addresses this need elegantly: by learning the distribution of normal surface appearance, the autoencoder produces a reconstruction that, when differenced against the defective input, directly highlights the anomalous region without requiring any spatial annotation in the training data [7].

1.2 Problem Statement

Despite the demonstrated classification effectiveness of HOG+LBP+SVM pipelines on clean datasets, two critical operational questions remain unresolved. First, to what extent does system performance degrade when test images are corrupted by Gaussian noise representative of industrial sensor degradation, and can a denoising preprocessing step effectively bridge this domain gap? Second,

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how can a purely classification-based system be extended to provide spatially meaningful defect localization without requiring expensive pixel-level annotations - a resource constraint common in industrial settings?

These questions define the scope of the present study. Existing robustness benchmarking work [8] has systematically documented how common corruptions degrade neural network classifiers, but comparable analysis for hand-crafted feature pipelines in industrial inspection remains limited [3]. Furthermore, most autoencoder-based anomaly localization studies apply fixed thresholds to difference maps [7], which performs poorly when reconstruction error magnitudes vary across images due to differences in casting geometry and background reflectance - a problem that motivates the adaptive thresholding strategy proposed here.

1.3 Research Objectives and Scope

This study pursues four primary objectives:

1. To quantify the robustness degradation of the HOG+LBP+SVM classification pipeline under controlled Gaussian noise injection (σ = 25), establishing a rigorous performance baseline and degradation profile.
2. To evaluate whether a two-stage Bilateral Filter + Non-Local Means (NL-Means) denoising pipeline, applied at inference time, can sufficiently recover classification performance on noisy inputs without sacrificing clean-data accuracy.
3. To assess model robustness against spatial distortions - simulated via random zoom-and-crop transformations - as a proxy for misaligned cameras or partial product visibility in real inspection setups.
4. To develop and evaluate an unsupervised defect localization module using a Convolutional Autoencoder, capable of producing spatially precise bounding boxes around anomalous regions using only image-level labels during training.

The scope is bounded to binary defect classification (defective vs. OK) and localization on the publicly available Real-Life Industrial Casting Product Dataset [1]. Deep learning-based end-to-end classifiers are explicitly excluded from Stage 1 to maintain methodological alignment with the hand-crafted feature extraction framework; CNN components appear exclusively in Stage 2 for localization.

1.4 Research Contributions

The principal contributions of this work are as follows:

1. A systematic five-scenario robustness evaluation framework for HOG+LBP+SVM-based casting defect detection, extending standard clean-test evaluation to include cross-domain Gaussian noise and spatial distortion scenarios.
2. A domain-bridging denoising strategy combining Bilateral Filter and NL-Means denoising that recovers near-baseline classification accuracy on noise-corrupted inputs, demonstrated empirically across all evaluation scenarios.
3. A zoom-and-crop training augmentation strategy that doubles effective training set size while specifically hardening models against spatial distortions, validated through Scenario 5 evaluation.
4. An adaptive percentile-based thresholding method for autoencoder reconstruction difference maps, replacing fixed thresholds with image-specific 93rd-percentile thresholds that robustly handle variable reconstruction error magnitudes across different casting samples.

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2. Methodology

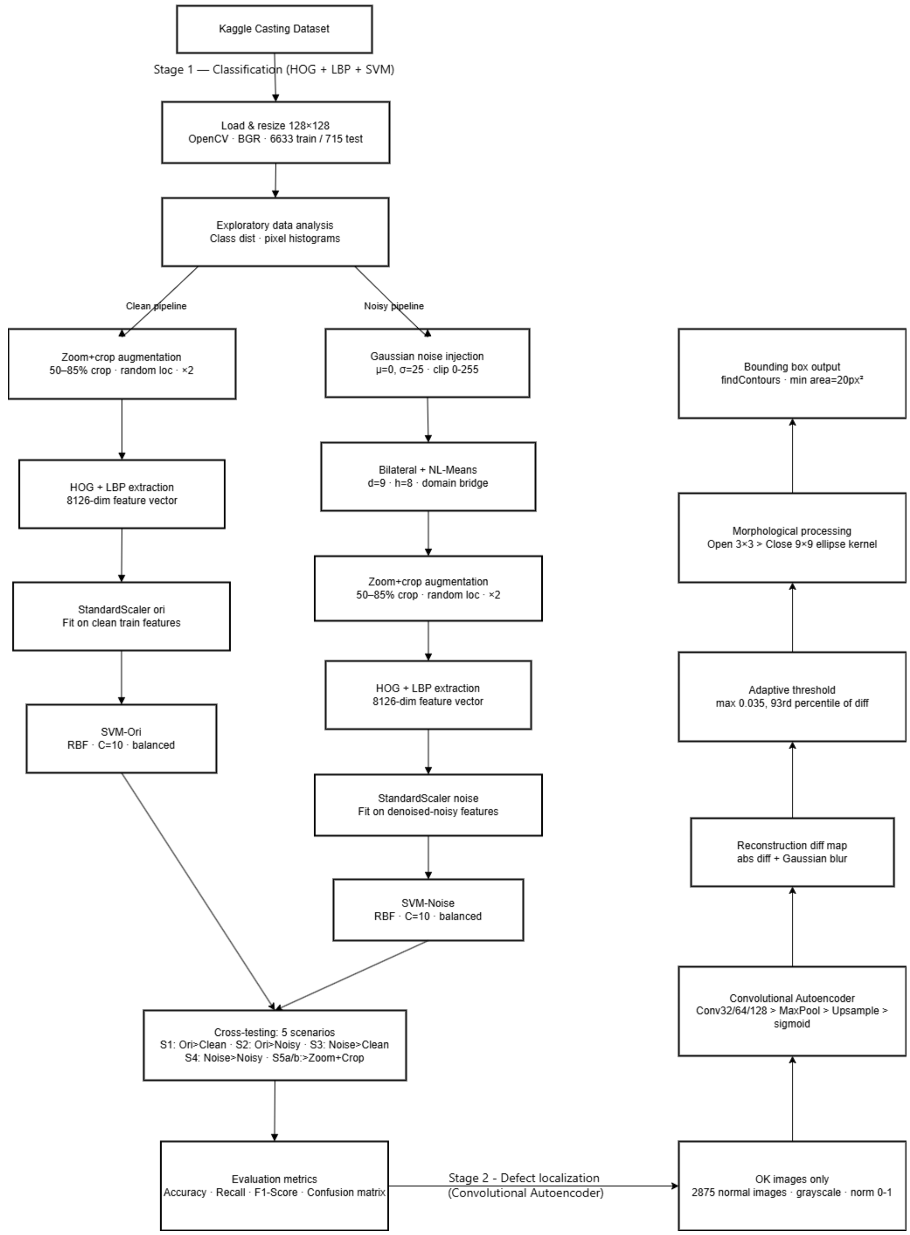

Figure 1. Complete Pipeline of the Proposed Two-stage Casting Defect Detection System.

The overall research pipeline is illustrated in Figure 1. The system comprises two sequential stages: Stage 1 performs binary defect classification through hand-crafted feature extraction and SVM-based learning across two parallel training pathways (clean and noise-augmented), while Stage 2 employs an unsupervised Convolutional Autoencoder to spatially localize defective regions within images predicted as defective by Stage 1. Each stage and its constituent components are described in detail in the subsections that follow.

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2.1 Dataset Description and Exploratory Data Analysis

The dataset used in this study is the Real-Life Industrial Casting Product Dataset, publicly available on Kaggle [1]. It consists of grayscale photographs of submersible pump impellers captured under controlled frontal-view conditions on a uniform background, collected directly from an active industrial production facility. This real-world provenance distinguishes it from laboratory-synthesized datasets, as it reflects genuine production variability including natural lighting inconsistencies and minor casting surface irregularities.

The dataset is partitioned into a training split of 6,633 images and a test split of 715 images, each containing two class categories: def_front (defective) and ok_front (normal). The training set comprises 3,758 defective images (56.7%) and 2,875 normal images (43.3%), while the test set contains 453 defective images (63.4%) and 262 normal images (36.6%). The mild class imbalance toward the defective class - slightly more pronounced in the test set - is addressed in the classifier configuration via the class_weight='balanced' parameter in the SVM. Exploratory analysis of pixel intensity distributions reveals that defective images exhibit heavier low-intensity tails, reflecting the dark shadow patterns characteristic of blow holes and surface inclusions, whereas normal images display a more uniform bell-shaped intensity distribution centered around mid-gray values. All images are loaded using OpenCV and uniformly resized to 128×128 pixels prior to any processing stage. Visual inspection of sample images confirms that defect class anomalies are primarily localized texture disruptions - irregular dark patches, pit clusters, and surface discontinuities - rather than global image-level differences, motivating the use of local feature descriptors (HOG, LBP) rather than global statistics for classification.

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2.2 Gaussian Noise Injection for Environmental Simulation

To simulate the progressive image quality degradation experienced by industrial camera sensors operating in harsh environments - including thermal noise, electromagnetic interference, and electronic readout noise - Gaussian noise is artificially injected into all images following the approach established in noise robustness benchmarking literature [8]. The injection process adds a noise matrix sampled from $N(\mu=0, \sigma=25)$ to each image channel, with pixel values subsequently clipped to the valid [0, 255] range to preserve valid image representation.

The standard deviation value of $\sigma = 25$ was selected as a compromise between realism and discriminability: it introduces sufficient structural perturbation to meaningfully degrade HOG gradient estimates - which are sensitive to pixel-level intensity differences - while preserving enough low-frequency image content to prevent complete information loss. This injection produces two parallel dataset versions (clean and noisy) from a single source, enabling controlled cross-condition evaluation. Importantly, noise injection is applied independently at inference time for each test scenario rather than once and cached, ensuring that evaluation reflects stochastic noise conditions rather than a fixed degraded copy.

2.3 Two-Stage Denoising Pipeline: Bilateral Filter and NL-Means

A core contribution of this study is the integration of a two-stage denoising pipeline as a preprocessing step within the noisy-data evaluation pathway. Rather than training models directly on noisy images and expecting them to internalize noise tolerance purely through exposure, the denoising pipeline explicitly reduces the domain gap between noisy input distributions and the clean feature distributions that the classifier was originally designed to handle.

The first stage applies a Bilateral Filter (kernel diameter $d = 9$, $\sigma\_color = 75$, $\sigma\_space = 75$), an edge-preserving smoothing operation that weights pixel contributions by both spatial proximity and

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intensity similarity. Unlike Gaussian blurring, which uniformly smooths across edges, the bilateral filter preserves gradient discontinuities at defect boundaries - critical for maintaining the integrity of HOG edge descriptors. The second stage applies Non-Local Means (NL-Means) denoising (h = 8, templateWindowSize = 7, searchWindowSize = 21), which exploits self-similarity across non-adjacent image patches to suppress residual noise that the bilateral filter leaves behind, particularly in smooth surface regions [9]. The combined pipeline achieves superior texture preservation compared to either filter alone, directly improving the consistency of LBP micro-texture descriptors extracted from denoised images. This denoising step is applied exclusively to noisy input images - both during the construction of the noisy training set (for SVM-Noise) and at test time for Scenarios 2, 4, 5b - and is intentionally omitted from clean-data pathways to avoid artificially blurring clean images. This design preserves the validity of clean-condition baselines while bridging the feature distribution gap for noisy conditions.

2.4 Hand-Crafted Feature Extraction: HOG and LBP

Feature extraction constitutes the representational core of Stage 1. Each image is first converted to grayscale, as both HOG and LBP operate on single-channel intensity representations. Two complementary descriptors are then computed and concatenated into a unified feature vector. The Histogram of Oriented Gradients (HOG) descriptor is computed with 9 orientation bins spanning 0°–180° (unsigned gradients), a cell size of 8×8 pixels, and 2×2 cell blocks with L2-Hys normalization. HOG captures the spatial distribution of gradient orientations across local image regions, making it well-suited for detecting structural anomalies such as blow holes and surface cracks that manifest as abrupt intensity changes [3]. The 128×128 input image with these parameters yields a HOG feature vector of length 8,100.

The Local Binary Pattern (LBP) descriptor employs a circular neighborhood with radius R = 3 and P = 24 sampling points in uniform mode [10]. For each pixel, the LBP code encodes the thresholded relationship between the central pixel and its neighbors, producing a rotation-invariant texture primitive. The resulting LBP map is summarized as a normalized histogram with P+2 = 26 bins, capturing the global micro-texture statistics of the image. LBP complements HOG by being sensitive to surface texture patterns - porosity pits, sand inclusions - that may produce negligible gradient responses but strong local texture signatures [4]. The final feature vector for each image is the concatenation of the HOG descriptor and the LBP histogram, yielding a combined representation of length 8,126.

2.5 Zoom-and-Crop Data Augmentation Strategy

A zoom-and-crop augmentation strategy is applied to the training set to improve model robustness against spatial distortions. For each of the 6,633 original training images, one augmented variant is generated by randomly cropping a sub-region covering 50–85% of the image dimensions at a randomly sampled location, then resizing the crop back to 128×128 pixels. This process effectively doubles the training set to 13,266 images (original + augmented) while preserving the original class label for each augmented sample.

This augmentation strategy serves two complementary purposes. First, it directly hardens both SVM models against the type of spatial distortion evaluated in Scenario 5, where test images are subjected to the same zoom-and-crop transformation. Second, it functions as a general regularization mechanism: by exposing the classifier to partial views of casting products, the learned decision boundary becomes less dependent on the global spatial layout of the image and more reliant on local texture and gradient patterns that are consistent across spatial scales. The augmented images for the noisy pipeline (SVM-Noise) additionally undergo Gaussian noise injection followed by Bilateral + NL-Means denoising, maintaining pipeline consistency with the non-augmented noisy training images.

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2.6 Feature Scaling and SVM Classifier Configuration

Prior to SVM training, all feature vectors are standardized using StandardScaler (zero mean, unit variance), fitted exclusively on the training set and applied consistently to all test sets. Standardization is critical for SVM with RBF kernel because the kernel computes Euclidean distances between feature vectors; features with large variance would otherwise dominate the distance computation and suppress the contribution of lower-variance descriptors [5]. Two independent scalers are maintained - one fitted on clean training features (scaler_ori) and one fitted on denoised-noisy training features (scaler_noise) - ensuring that each model's test features are scaled consistently with its training distribution.

Two SVM classifiers are trained with identical hyperparameters: RBF kernel, regularization parameter C = 10, kernel bandwidth parameter γ = 'scale' (computed as 1/(n_features × Var(X))), and class_weight = 'balanced'. The SVM-Ori model is trained on clean feature representations (augmented training set), serving as the baseline classifier. The SVM-Noise model is trained on denoised-noisy feature representations (same augmented set, processed through the noise+denoising pipeline), serving as the noise-hardened alternative. The class_weight='balanced' parameter adjusts class weights inversely proportional to class frequencies, preventing the mild defective-class majority from biasing the SVM decision boundary toward predicting defective for ambiguous feature vectors - a failure mode observed without this correction in preliminary experiments.

Overfitting is assessed by comparing training accuracy against test accuracy for both models. Both models achieve 100% training accuracy, which is expected for SVM with C = 10 on a separable feature space. The gap between training and test accuracy (SVM-Ori: 99.16% test accuracy; SVM-Noise: 98.60% test accuracy on their respective aligned test sets) is small, confirming that both models generalize well and that the training accuracy does not reflect memorization of noise-specific patterns.

2.7 Convolutional Autoencoder Architecture and Training Protocol

The Stage 2 defect localization module is built upon a Convolutional Autoencoder (CAE), which learns the visual distribution of non-defective casting surfaces in a fully unsupervised manner. The fundamental operating principle of the module exploits an asymmetry in reconstruction fidelity: an autoencoder trained exclusively on normal images learns to faithfully reconstruct normal surface appearances, while defective regions containing texture anomalies absent from the training distribution produce elevated pixel-wise reconstruction errors that can be spatially localized [13].

The training set for this module consists exclusively of the 2,875 normal (ok_front) images from the training split. All images are converted to single-channel grayscale and normalized to the continuous range [0, 1] prior to processing. This normalization ensures numerical stability during gradient-based optimization and constrains the final sigmoid output layer to the same unit range.

The encoder pathway comprises three sequential convolutional blocks. The first block applies a Conv2D layer with 32 filters of size 3×3, ReLU activation, and same-padding to maintain spatial dimensions, followed by a MaxPooling2D operation with pool size 2×2 and same-padding, reducing the feature map from 128×128 to 64×64 pixels. The second block applies 64 filters with identical configuration, further reducing the representation to 32×32 pixels. The third block applies 128 filters and produces the compressed latent representation at 16×16 pixels. The resulting encoded feature map captures the statistical regularities of normal casting surfaces in a compact, high-level representation that the decoder must reconstruct.

The decoder mirrors the encoder in reverse. Three Conv2DTranspose layers with filter counts of 128, 64, and 32, each followed by an UpSampling2D operation with factor 2×2, progressively restore spatial resolution from 16×16 back to the original 128×128 pixels. The final reconstruction layer applies a single Conv2D filter with a 3×3 kernel and sigmoid activation, producing a single-channel output

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image in the [0, 1] range.
The model is compiled with the Adam optimizer and trained to minimize pixel-wise mean squared error (MSE) between each input normal image and its reconstruction. Training proceeds for 30 epochs with a batch size of 32. The choice of MSE as the reconstruction loss is appropriate for grayscale surface imagery because pixel-wise squared differences effectively penalize reconstruction deviations at individual spatial locations, directly linking training optimization to the spatial reconstruction quality that the localization step depends upon [14].

2.8 Defect Localization: Adaptive Thresholding and Morphological Processing

Following Stage 1 classification, all test images predicted as defective are passed to the Stage 2 localization pipeline. The localization procedure transforms the raw reconstruction difference between the autoencoder input and output into spatially precise bounding box annotations through four sequential processing steps.

1. **Reconstruction and Difference Map Computation**. The input image, preprocessed to grayscale and normalized to [0, 1], is passed through the trained autoencoder to obtain a reconstructed image representing how the casting surface would appear if all defects were absent. The pixel-wise absolute difference between the input and the reconstruction is computed, yielding an unsigned difference map where high values indicate pixels that deviate substantially from the learned normal surface model. A Gaussian blur is then applied to this map to suppress isolated single-pixel deviations arising from stochastic reconstruction imprecision, producing a spatially smooth anomaly score map.
2. **Adaptive Percentile-Based Thresholding**. Rather than applying a fixed global threshold, which would perform inconsistently across images with differing absolute reconstruction error magnitudes due to casting geometry and surface reflectance variability, an image-specific adaptive threshold is computed for each sample. The threshold is defined as the maximum of two values: a global floor value of 0.035, which prevents spuriously low thresholds on very uniform castings, and the 93rd percentile of all pixel values within the current smoothed difference map. This percentile-based formulation ensures that only the top 7% of highest-error pixels in each individual image exceed the threshold, robustly isolating the most anomalous spatial regions regardless of the overall reconstruction error scale. Pixels exceeding this adaptive threshold are set to one in a binary anomaly mask, while all remaining pixels are set to zero.
3. **Morphological Refinement**. The binary mask is refined through two morphological operations. First, an opening operation using a 3×3 elliptical structuring element is applied to remove isolated noise pixels and small spurious high-error regions that do not correspond to physically meaningful defect areas. Second, a closing operation using a 9×9 elliptical structuring element fills internal gaps within candidate defect regions, producing smoother and more spatially contiguous anomaly masks that better reflect the physical extent of surface defects. The elliptical kernel shape is chosen to align with the curved boundary morphology typical of casting surface anomalies such as blow holes and pit clusters.
4. **Contour Extraction and Bounding Box Generation**. Connected component analysis is performed on the refined binary mask using the findContours algorithm. Connected regions with a total pixel area below 20 pixels are discarded as residual noise responses too small to represent meaningful surface defects. Each surviving connected region is enclosed within an axis-aligned bounding box, which is overlaid on the original image to produce the final annotated localization output.

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2.9 Cross-Domain Evaluation Scenarios and Performance Metrics

A structured five-scenario evaluation framework is established to comprehensively assess pipeline robustness across distinct combinations of model training condition and test data condition. Each scenario isolates a specific axis of domain variation, enabling attribution of performance differences to clearly defined factors.

Scenario 1 (Original Model, Clean Test Data) serves as the baseline, evaluating the SVM-Ori model on the unmodified 715-image test set. This scenario establishes the performance ceiling of the pipeline under ideal matched conditions and provides the reference point against which all cross-condition degradations are measured.

Scenario 2 (Original Model, Noisy+Denoised Test Data) introduces a deliberate domain mismatch by evaluating the SVM-Ori model on test images corrupted with Gaussian noise (sigma = 25) and subsequently processed through the Bilateral+NL-Means denoising pipeline. This scenario quantifies the performance degradation suffered by a production system trained on clean data but deployed in a noisy sensor environment.

Scenario 3 (Noisy Model, Clean Test Data) evaluates the reverse direction of domain mismatch by applying the SVM-Noise model to clean test images. This scenario measures whether noise-hardened training introduces any performance regression on clean inputs, characterizing the cost of noise adaptation.

Scenario 4 (Noisy Model, Noisy+Denoised Test Data) establishes the noise-aligned performance ceiling by evaluating the SVM-Noise model on its matching noisy-then-denoised test set. This represents the ideal deployment configuration for a noise-exposed system and demonstrates the effectiveness of domain-matched training and inference.

Scenario 5a (Original Model, Zoom+Crop Test Data) and Scenario 5b (Noisy Model, Zoom+Crop Test Data) evaluate spatial distortion robustness on a dedicated zoom-and-crop test set of 100 images comprising 64 defective and 36 OK samples, generated by applying random 50 to 85% zoom-and-crop transformations.

Three metrics are reported for all classification scenarios: overall accuracy, defective-class recall (sensitivity), and defective-class F1-score. Confusion matrices are generated for each scenario, reporting true positives (correctly identified defective), true negatives (correctly identified OK), false positives (OK misclassified as defective), and false negatives (defective misclassified as OK). For the Stage 2 localization module, performance is reported as the proportion of defective images receiving at least one bounding box annotation, the average number of bounding boxes generated per image, and the maximum bounding box count across all evaluated samples.

3. Results and Analysis

3.1 Exploratory Data Analysis: Class Distribution and Image Statistics

The dataset comprises 7,348 images in total across training and test splits. The training set contains 6,633 images distributed as 3,758 defective samples (56.7%) and 2,875 normal samples (43.3%). The test set contains 715 images with 453 defective samples (63.4%) and 262 normal samples (36.6%). The mild class imbalance - consistently skewed toward the defective class - is more pronounced in the test set than in the training set, which motivated the use of class_weight='balanced' in both SVM models to prevent the classifier from defaulting to the majority class under distribution shift.

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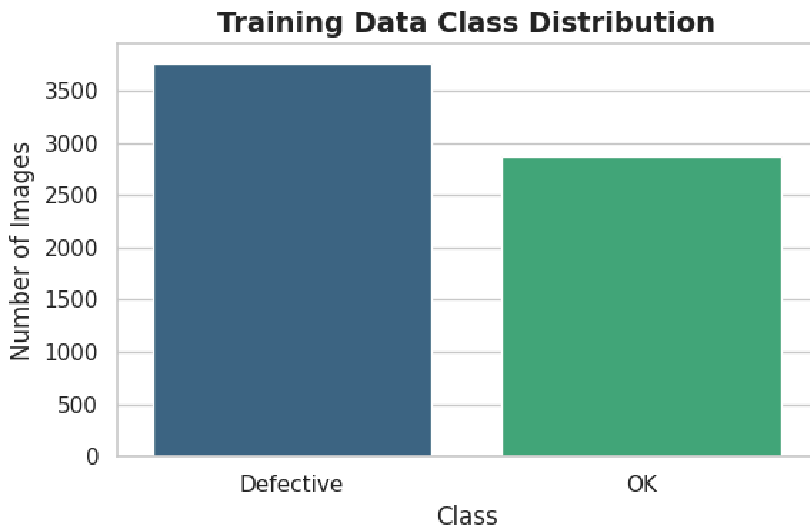

Figure 2. Class Distribution of the Training Set

Visual inspection of sample images reveals that defective castings display spatially localized anomalies - irregular dark patches, pit clusters, and surface discontinuities typically confined to one region of the product surface - rather than global image-level differences. Normal castings, by contrast, present smooth and visually uniform surfaces with consistent reflectance patterns. This spatial locality of defects directly motivates the use of local feature descriptors (HOG and LBP) rather than global statistics, and foreshadows the effectiveness of the autoencoder-based localization approach in Stage 2.

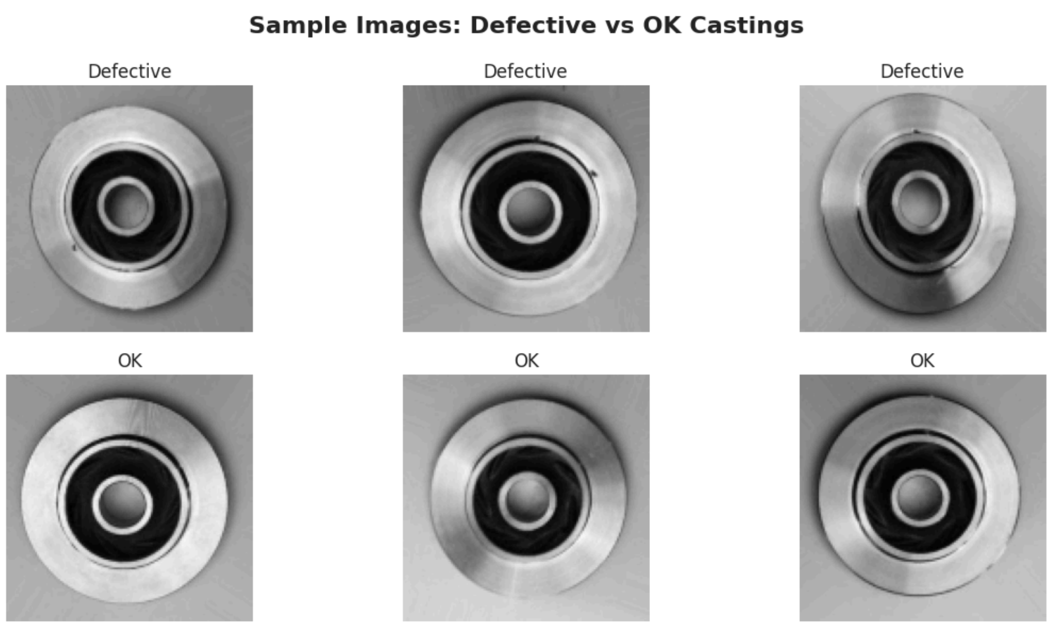

Figure 3. Representative Samples From the Training Set

Pixel intensity distribution analysis further confirms this distinction. Defective images exhibit

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heavier low-intensity tails, reflecting the dark shadow patterns characteristic of blow holes and surface voids. Normal images display a more symmetric, bell-shaped intensity distribution centered around mid-gray values. This distributional difference in the raw pixel space provides initial evidence that discriminative texture and gradient features should be learnable by the HOG+LBP pipeline.

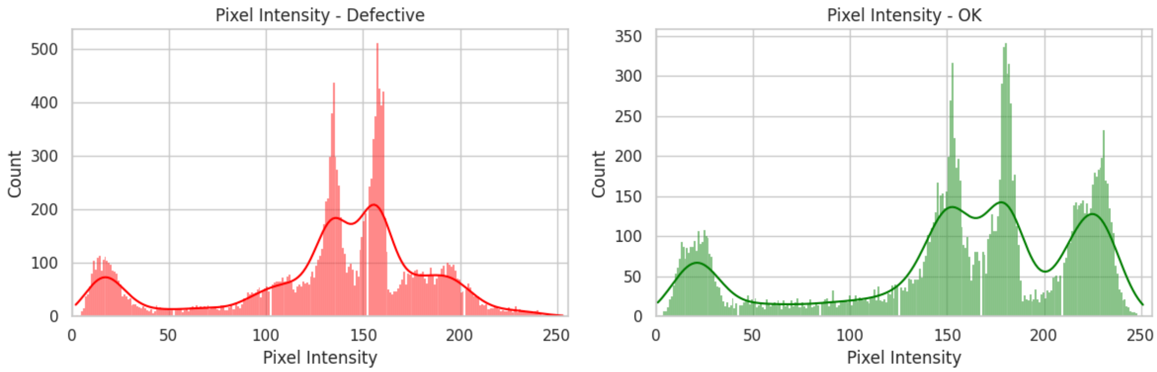

Figure 4. Pixel Intensity Histograms for a Representative Defective Image (Left) and Normal Image (Right)

3.2 Noise Injection and Denoising Effectiveness

Gaussian noise with parameters μ = 0 and σ = 25 was injected into all images to simulate industrial sensor degradation. The qualitative effect is clearly visible: high-frequency speckle noise overlays the image uniformly, disrupting the smooth surface gradients of normal castings and adding spurious edge responses throughout defective regions.

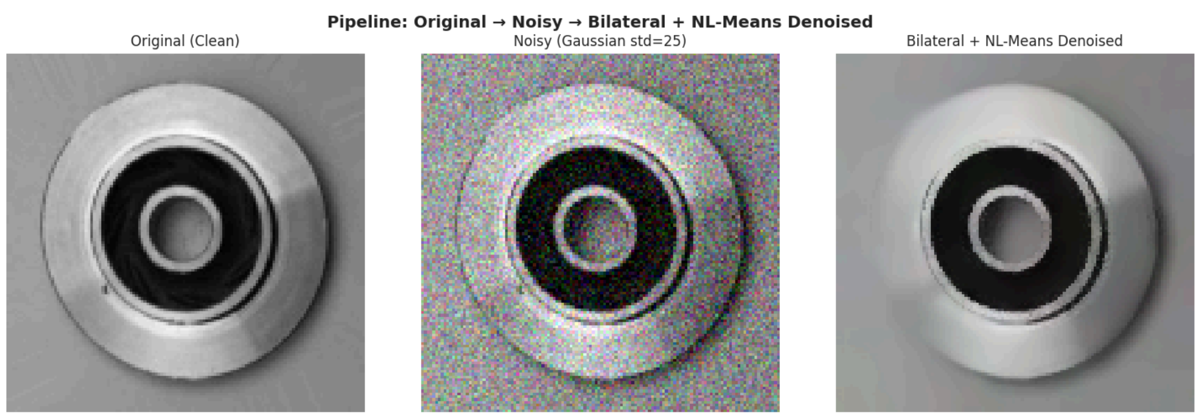

Figure 5. Visual Effect of the Noise Injection and Denoising Pipeline on a Representative Casting Image

The two-stage denoising pipeline - Bilateral Filter (d=9, σ_color=75, σ_space=75) followed by NL-Means (h=8, templateWindowSize=7, searchWindowSize=21) - recovers a visually clean representation that closely approximates the original. Crucially, the bilateral filter's edge-preserving property ensures that the sharp boundaries at defect sites are not blurred away, while NL-Means suppresses the residual fine-grained speckle that the bilateral filter leaves behind. The denoised image

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retains the structural and textural information necessary for both HOG gradient computation and LBP micro-texture encoding.

It is important to note, however, that denoising does not perfectly reconstruct the original image. Subtle differences in the feature distribution between clean and denoised-noisy images persist, as reflected in the performance gap between Scenarios 1 and 2, and between Scenarios 3 and 4 (discussed further in Section 3.4). The denoising pipeline functions as a domain-bridging mechanism rather than a perfect noise reversal.

3.3 SVM Training Performance and Overfitting Analysis

Both SVM models were trained on the augmented training set comprising 13,266 images - the 6,633 original images plus 6,633 zoom-and-crop variants. Training and test accuracies for both models are summarized in Table 1.

Table 1. Overfitting Analysis: Training Accuracy vs. Test Accuracy for Both SVM Models on Their Respective Aligned Test Sets.

Model	Training Accuracy (%)	Test Accuracy (%)
SVM-Ori (Clean)	100.00	99.16
SVM-Noise (Denoised-Noisy)	100.00	98.60

Both models achieve 100% training accuracy, which is expected behavior for an SVM with C=10 operating on a high-dimensional, well-separated feature space. The small gap between training and aligned test accuracy - 0.84 percentage points for SVM-Ori and 1.40 points for SVM-Noise - confirms that neither model has overfit to its training distribution. The slightly larger gap in SVM-Noise is consistent with the inherently higher variance introduced by stochastic noise injection during feature extraction: each training pass through extract_features_noisy generates a different noise realization, meaning the model sees a diverse sampling of denoised-noisy feature representations rather than fixed training points, which acts as an implicit regularizer but also slightly increases estimation variance at test time.

The zoom-and-crop augmentation contributed meaningfully to the training set diversity. By doubling the effective training size and exposing both models to partial spatial views of casting products, the augmentation strategy directly serves the robustness requirements tested in Scenario 5, where models are evaluated on images that have never been seen in their full-frame form. The class label consistency between original and augmented samples is valid because defects are surface-level anomalies that are preserved under cropping - a crop of a defective casting will still contain defect texture, while a crop of a normal casting will present normal surface texture, maintaining the discriminative signal in both classes.

3.4 Robustness Evaluation under Gaussian Noise: Scenarios 1 through 4

The classification results across the four noise-related evaluation scenarios are presented in Table 2 and the corresponding confusion matrices in Figures 6 through 9.

Table 2. Classification Performance Across Four Evaluation Scenarios

Scenario	Accuracy (%)	Recall (%)	F1-Score (%)

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1 - Original Model, Clean Test Data	99.16	98.68	99.33
2 - Original Model, Noisy+Denoised Test Data	60.84	38.19	55.27
3 - Noisy Model, Clean Test Data	71.19	60.04	72.53
4 - Noisy Model, Noisy+Denoised Test Data	98.60	98.45	98.89

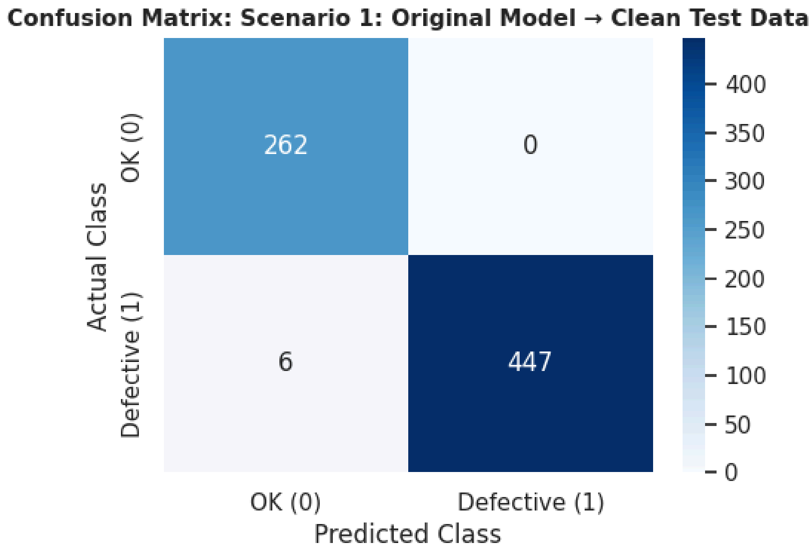

Figure 6. Confusion Matrix of Scenario 1

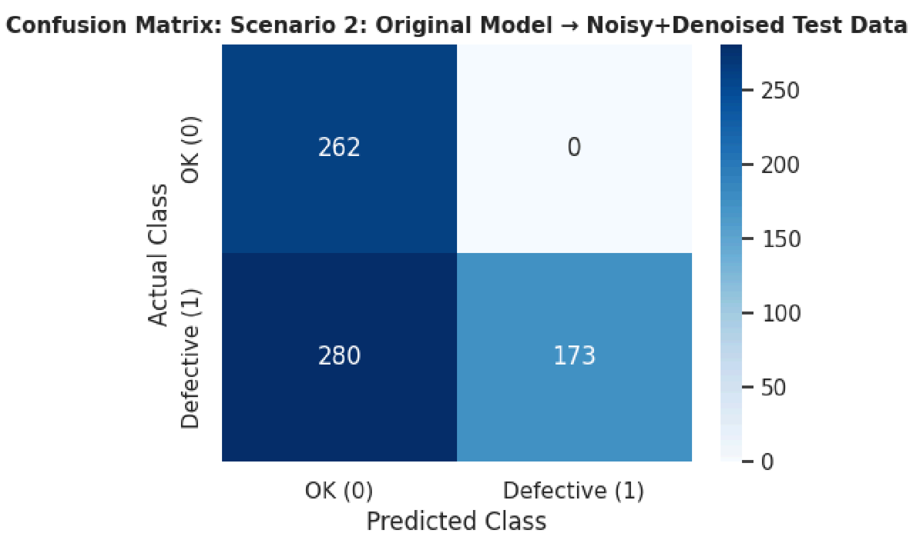

Figure 7. Confusion Matrix of Scenario 2

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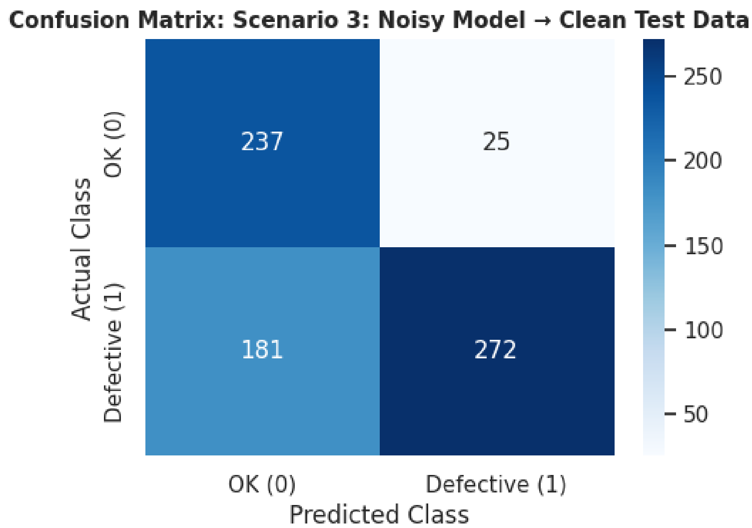

Figure 8. Confusion Matrix of Scenario 3

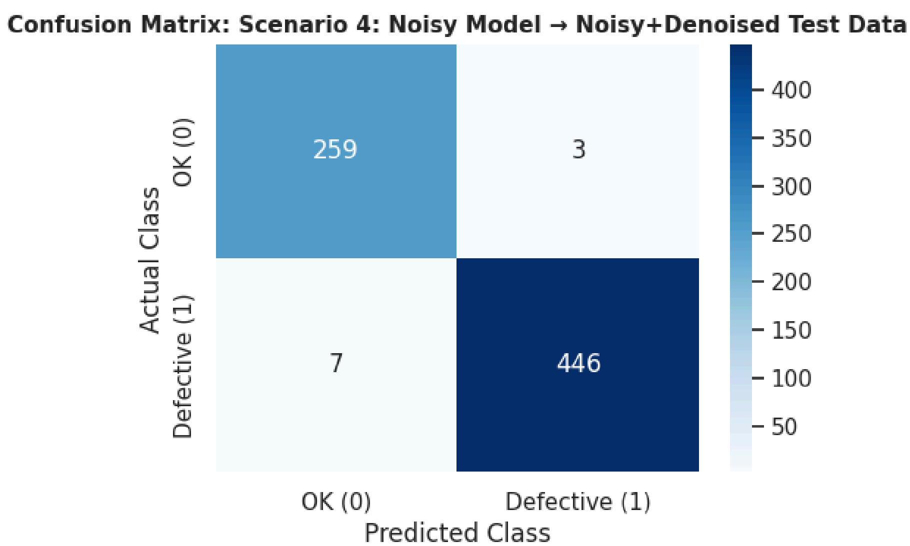

Figure 9. Confusion Matrix of Scenario 4

Scenario 1 establishes the performance ceiling under matched clean conditions. The SVM-Ori model achieves 99.16% accuracy, 98.68% recall, and 99.33% F1-score across the full 715-image test set. As shown in Figure 6, the confusion matrix reveals only 6 false negatives (defective images misclassified as OK) and zero false positives, confirming near-perfect separation between the defective and normal classes when training and testing distributions are fully aligned. This result validates the discriminative power of the 8,126-dimensional HOG+LBP feature representation for clean casting imagery.

Scenario 2 exposes the most critical vulnerability of the pipeline. When the same SVM-Ori model is evaluated on noisy-then-denoised test images, accuracy collapses to 60.84%, recall falls drastically to 38.19%, and F1-score drops to 55.27%. As shown in Figure 7, the confusion matrix reveals that 280 of

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453 defective images (61.8%) are misclassified as normal, while all 262 normal images remain correctly classified (zero false positives). This strongly asymmetric error profile reveals the mechanism of failure: Gaussian noise corrupts the precise gradient magnitude and orientation information on which HOG descriptors depend, and while the bilateral-NL-Means denoising substantially attenuates the noise, the residual feature distribution shift between clean training features and denoised-noisy test features is sufficient to render defective surface signatures unrecognizable to the clean-trained classifier. The intact specificity for normal images indicates that the classifier retains a firm representation of normal surface appearance, but the noise-altered texture features of defective surfaces drift into the normal-class decision region after denoising.

Scenario 3 characterizes the reverse cross-condition scenario. The SVM-Noise model evaluated on clean test images achieves 71.19% accuracy and 60.04% recall, with an F1-score of 72.53%. As shown in Figure 8, the confusion matrix shows 181 false negatives and 25 false positives across the 715 test images. Compared to Scenario 2, the magnitude of degradation is substantially smaller (71.19% versus 60.84%), indicating an asymmetry in cross-condition generalization: the denoised-noisy feature distribution partially overlaps with the clean feature distribution, enabling partial generalization. Nonetheless, the noise-hardened model still loses considerable sensitivity when applied to clean data, confirming that training on denoised-noisy features shifts the decision boundary in ways that are not fully compatible with clean-image feature representations.

Scenario 4 demonstrates the near-complete recovery achievable through domain-matched deployment. When the SVM-Noise model is evaluated on its aligned noisy-then-denoised test set, accuracy rises to 98.60%, recall to 98.45%, and F1-score to 98.89%. As shown in Figure 9, the confusion matrix reports only 7 false negatives and 3 false positives, achieving near-parity with the Scenario 1 baseline. The residual gap of 0.56 percentage points in accuracy relative to Scenario 1 reflects the irreducible feature distribution shift introduced by imperfect noise cancellation, not a limitation of the SVM classifier. This result validates the central claim of the noise-hardening strategy: when training preprocessing and inference preprocessing are aligned, the domain gap introduced by sensor noise can be effectively bridged without architectural modifications to the classification model.

The results across Scenarios 1 through 4 collectively demonstrate a clear principle: performance is governed more by the alignment between training and test feature distributions than by the absolute presence of noise in the test data. A mismatched clean model degrades severely on noisy data (Scenario 2), while a noise-hardened model can closely match clean-baseline performance when evaluated on its matched test distribution (Scenario 4).

3.5 Robustness Evaluation under Spatial Distortion: Scenario 5

Figure 10. Samples Random Zoom+Crop Images of Scenario 5

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Table 3. Classification Performance Across Scenario 5a and 5b

Scenario	Accuracy (%)	Recall (%)	F1-Score (%)
5a - Original Model, Zoom+Crop Test Data	76.00	81.25	81.25
5b - Noisy Model, Zoom+Crop Test Data	76.00	78.12	80.65

Scenarios 5a and 5b evaluate pipeline robustness against spatial distortions using a dedicated zoom-and-crop test set of 100 images comprising 64 defective and 36 OK samples. The model robustness summary across all scenarios is visualized in Figure 11.

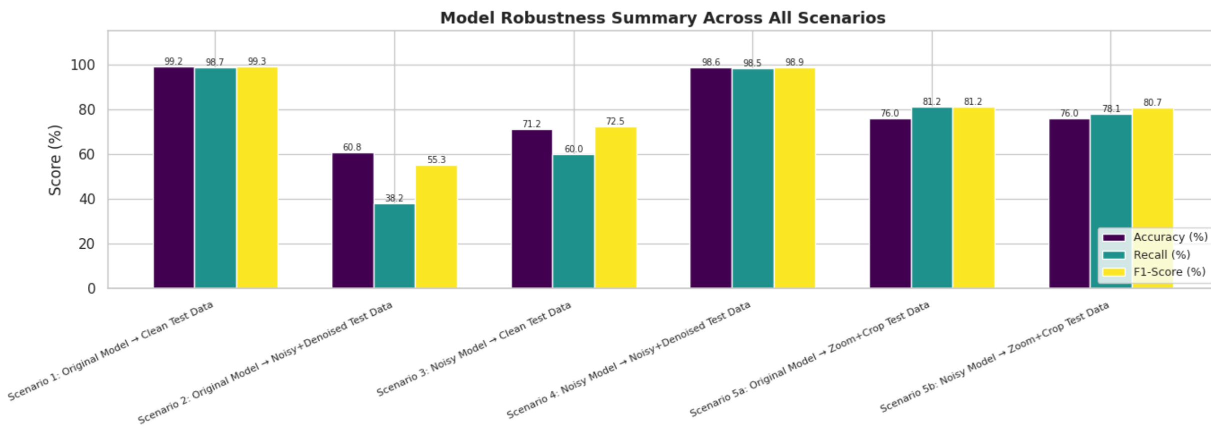

Figure 11. Model Robustness Summary Across All Scenarios

Scenario 5a evaluates the SVM-Ori model against the zoom-and-crop test set. The model achieves 76.00% accuracy, 81.25% recall, and 81.25% F1-score.

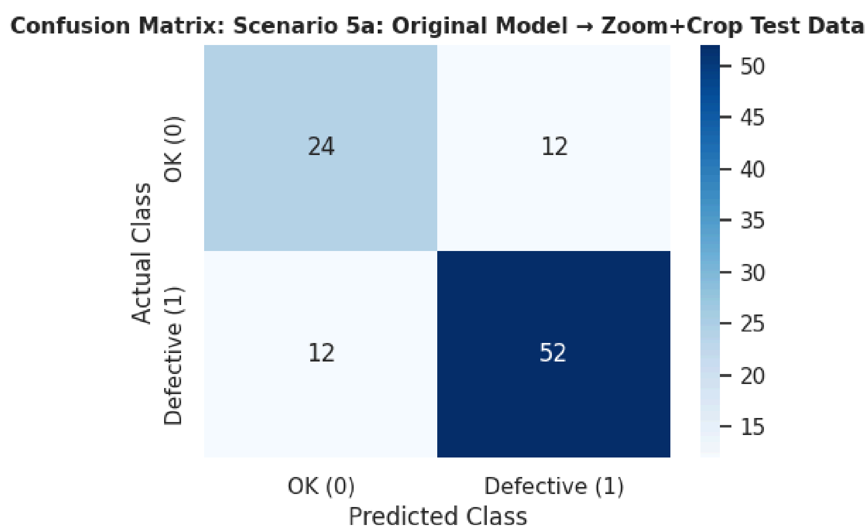

Figure 12. Confusion of Matrix Scenario 5a

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As shown in Figure 12, the confusion matrix reports 52 true positives, 24 true negatives, 12 false positives, and 12 false negatives. The symmetric distribution of misclassifications between the two error types is notable: the model misidentifies 12 OK samples as defective and simultaneously fails to detect 12 genuinely defective samples. This pattern suggests that when zoom-and-crop transformations expose unusual or partial surface regions, feature vectors from both classes shift toward ambiguous regions of the decision space, producing balanced error rates rather than the class-specific failure observed in the noise scenarios.

Scenario 5b evaluates the SVM-Noise model under the same spatial distortion. The model achieves an identical accuracy of 76.00%, with a slightly lower recall of 78.12% and F1-score of 80.65%.

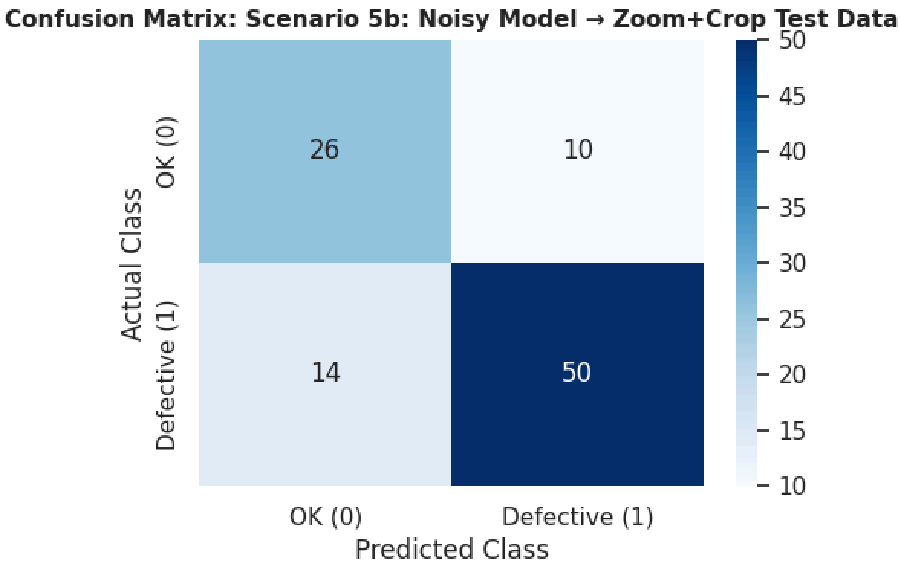

Figure 13. Confusion Matrix of Scenario 5b

As shown in Figure 13, the confusion matrix reports 50 true positives, 26 true negatives, 10 false positives, and 14 false negatives. The noise-hardened model demonstrates marginally improved specificity relative to Scenario 5a (10 false positives versus 12), but slightly reduced sensitivity (14 false negatives versus 12), indicating a minor decision boundary shift introduced by noise-aware training that slightly favors normal-class predictions.

The convergence of both models to identical accuracy under spatial distortion (76.00%) indicates that zoom-and-crop training augmentation, which was uniformly applied to both SVM-Ori and SVM-Noise during training, is the primary driver of spatial robustness. Neither model holds a meaningful advantage over the other in this scenario. However, both models fall substantially below their matched-condition performance (99.16% and 98.60% respectively), confirming that spatial distortions representing approximately 15 to 50% partial views of the casting surface constitute a distinct and significant challenge for the HOG+LBP feature pipeline. This degradation arises because HOG and LBP descriptors computed on a cropped sub-region capture a different spatial distribution of gradients and textures than those computed on the complete casting surface, reducing the discriminative alignment between training and test feature distributions despite the augmentation strategy.

3.6 Autoencoder Anomaly Detection and Defect Localization Results

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The Stage 2 Convolutional Autoencoder localization module was evaluated on 200 defective images drawn from the original test set. The autoencoder, trained exclusively on the 2,875 normal casting images, produces reconstruction-based difference maps that spatially highlight regions where the defective input deviates from the learned normal surface model.

Localization performance is summarized as follows. Of the 200 evaluated defective images, all 200 received at least one bounding box annotation, yielding a detection coverage rate of 100.0%. No defective images produced an empty output (zero missed detections). The average number of bounding boxes generated per image is 5.91, reflecting the tendency of the pipeline to resolve individual casting defect regions into multiple spatially distinct bounding box proposals. The maximum bounding box count observed across the evaluation set is 13, recorded for samples where defects are distributed across multiple spatially separated surface sub-regions.

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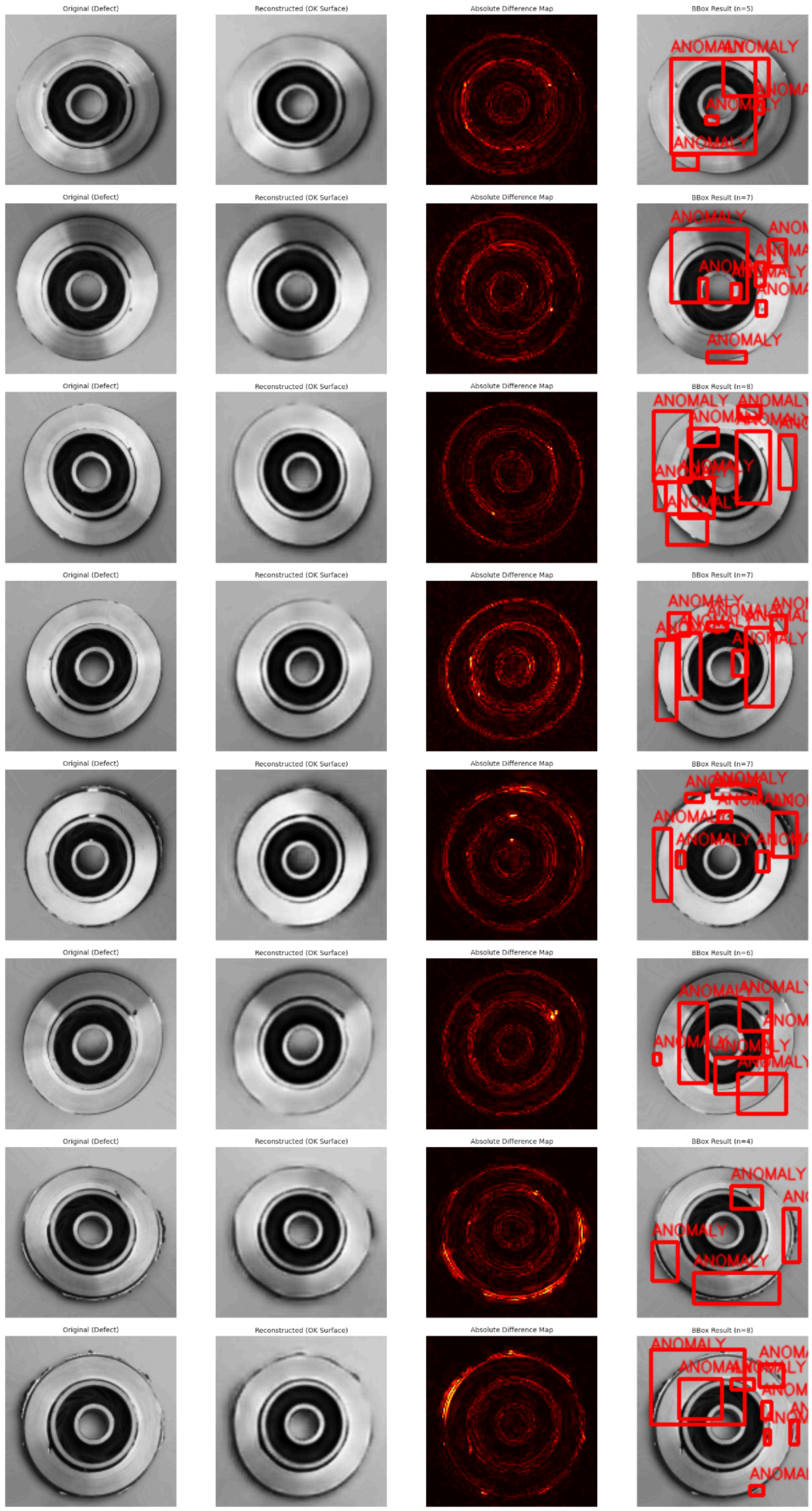

Figure 14. Autoencoder Anomaly Detection Results

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Figure 14 presents representative localization outputs for eight defective samples. Each row displays four panels: the original defective casting image, the autoencoder reconstruction representing the inferred normal surface, the absolute reconstruction difference map, and the final bounding box output after adaptive thresholding and morphological post-processing. Several consistent patterns are observable across all displayed samples. The reconstructed images present smooth, visually uniform surface appearances that suppress the irregular dark patches and pit clusters visible in the defective originals. The difference maps show elevated reconstruction error concentrated at the locations of these surface anomalies, while regions of normal casting surface produce low-magnitude, spatially uniform difference responses. This spatial specificity of reconstruction error confirms that the autoencoder has successfully internalized a statistical model of normal surface appearance rather than merely learning a general image smoothing function.

The adaptive 93rd-percentile threshold effectively isolates the anomalous high-error regions in each image independently of the absolute reconstruction error scale. The bounding box counts across the eight illustrated samples range from 4 to 8, consistent with the dataset-wide average of 5.91, indicating that multiple localized anomaly detections per image are the norm rather than the exception. Morphological post-processing visibly consolidates fragmented high-error regions into coherent bounding boxes while suppressing small spurious detections that are not supported by sustained reconstruction error over a meaningful area.

3.7 Cross-Stage Comparative Analysis and Discussion

The combined results across all evaluation scenarios reveal several important patterns about the robustness characteristics of the proposed two-stage pipeline that deserve systematic discussion.
The domain shift sensitivity of HOG+LBP features. The most consequential finding is the extreme sensitivity of the SVM-Ori model to feature distribution shift induced by Gaussian noise. The 38.32 percentage point accuracy collapse from Scenario 1 (99.16%) to Scenario 2 (60.84%) is entirely attributable to the mismatch between clean training features and denoised-noisy test features. This vulnerability is intrinsic to the HOG descriptor architecture: because HOG fundamentally computes local gradient magnitudes and orientations at the pixel level, Gaussian noise introduces dense spurious gradient responses uniformly across all image regions. Although the bilateral-NL-Means pipeline substantially attenuates these responses, residual distribution shifts persist that the clean-trained SVM boundary cannot accommodate. The near-zero false positive rate in Scenario 2 (TN = 262, FP = 0) reveals that the classifier retains a firm representation of normal surface appearance but loses sensitivity to defect signatures that have been subtly altered by the denoising process. This finding implies that in real deployment, a clean-data-trained classifier should never be exposed to noise-processed inputs without retraining under matching conditions.

The effectiveness and cost of noise-hardened training. Scenario 4 demonstrates a recovery of approximately 37.76 percentage points relative to the cross-condition baseline of Scenario 2, reaching 98.60% accuracy with matched preprocessing. This near-complete recovery at a residual cost of only 0.56 percentage points relative to the clean baseline (Scenario 1) confirms that the domain-bridging denoising strategy is both effective and efficient. The small residual gap reflects irreducible feature distribution differences that denoising cannot fully eliminate, not an architectural limitation of the SVM classifier. Scenario 3 (71.19% accuracy, SVM-Noise on clean data) reveals the cost of noise adaptation: some degradation in clean-condition performance occurs when the model is trained on denoised-noisy features rather than clean features. However, this cost is substantially smaller than the degradation in the opposite direction (71.19% versus 60.84%), suggesting that the noise-hardened model is a more conservative choice for deployment environments where noise conditions may vary, as it maintains reasonable performance across both clean and noisy conditions while the clean model fails badly under noise.

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The role of augmentation in spatial robustness. The convergence of both models to identical accuracy under spatial distortion (76.00%) demonstrates that zoom-and-crop training augmentation is the primary determinant of spatial robustness in this pipeline. Neither noise-hardening nor clean training confers a meaningful advantage under partial-view conditions, suggesting that the augmentation strategy effectively equalizes the spatial robustness of both models. The 23 percentage point gap between matched-condition performance and spatial-distortion performance (approximately 99% versus 76%) indicates that partial spatial views present a fundamentally different challenge from Gaussian noise: where noise robustness can be substantially recovered through domain-matched training, spatial distortions expose limitations in the global feature representation that augmentation partially but incompletely addresses.

The complementary role of Stage 2 localization. The 100% detection coverage of the Stage 2 autoencoder across all 200 evaluated defective samples represents a substantive contribution over the binary Stage 1 output. While Stage 1 produces a class label that is operationally actionable (reject or pass), it provides no information about where on the casting surface the defect is located. The Stage 2 module delivers this spatial information without requiring any pixel-level annotation during training, relying only on the image-level class labels available from the original dataset. The average of 5.91 bounding boxes per defective image reflects the multi-site nature of many casting defects rather than a precision failure: castings with multiple spatially separated surface anomalies legitimately generate multiple detection regions. The adaptive thresholding strategy, by using image-specific 93rd-percentile thresholds, successfully handles the variability in reconstruction error magnitude across different casting geometries without producing catastrophic over-detection on uniform-surface regions or under-detection on high-variance surfaces.

4. Conclusion and Recommendations

4.1 Summary of Research Findings

This study presented and evaluated a two-stage pipeline for casting defect classification and localization on the Real-Life Industrial Casting Product Dataset, with a systematic focus on robustness under Gaussian noise and spatial distortions. The principal findings are consolidated below.

The HOG+LBP+SVM pipeline achieves near-perfect classification performance (99.16% accuracy, 99.33% F1-score) under matched clean training and testing conditions, confirming the effectiveness of hand-crafted feature descriptors for industrial defect classification when training and deployment conditions are aligned.

Gaussian sensor noise introduces a severe domain shift that reduces accuracy from 99.16% to 60.84% when a clean-trained model is evaluated on noisy-then-denoised inputs without preprocessing alignment. The primary failure mode is the systematic misclassification of defective images as normal (recall drops to 38.19%), caused by noise-induced corruption of the gradient-based features on which HOG descriptors depend.

A domain-bridging strategy combining Bilateral Filter and Non-Local Means denoising preprocessing with a dedicated noise-hardened SVM model effectively recovers classification performance to 98.60% accuracy under noisy conditions, representing a recovery of 37.76 percentage points. This near-baseline recovery is achieved without any architectural modification to the classifier, demonstrating that preprocessing alignment is a practical and sufficient mechanism for noise robustness in hand-crafted feature pipelines.

Both models demonstrate comparable spatial distortion robustness (76.00% accuracy, F1-scores above 80%) when evaluated on zoom-and-crop distorted test data, confirming that the zoom-and-crop training augmentation strategy is the primary contributor to spatial robustness and that its benefits are consistent across both training configurations.

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The Stage 2 Convolutional Autoencoder achieves complete localization coverage on all 200 evaluated defective samples (100% detection rate), generating an average of 5.91 bounding boxes per image with a maximum of 13, without requiring any pixel-level spatial annotation during training. The adaptive percentile-based thresholding and morphological post-processing pipeline produces spatially coherent bounding box annotations that closely correspond to visible surface anomalies across diverse casting samples.

4.2 Limitations of the Proposed Approach

Several limitations constrain the scope and generalizability of the proposed pipeline and must be acknowledged when interpreting the reported results.

First, the robustness evaluation is conducted at a single noise level of sigma = 25. Real industrial environments exhibit noise with substantially variable standard deviations and may include non-Gaussian noise types such as salt-and-pepper noise, multiplicative speckle noise, or structured electromagnetic interference patterns. The performance characteristics of both the denoising pipeline and the SVM-Noise model under substantially different noise profiles remain uncharacterized in this study.

Second, the spatial distortion robustness assessment is limited to random zoom-and-crop transformations covering 50 to 85% of the original image. Other practically relevant spatial degradations such as rotation, perspective distortion, partial occlusion, or motion blur are not evaluated. The pipeline may exhibit different spatial robustness characteristics under these alternative distortion types.

Third, the Stage 2 autoencoder produces multiple bounding boxes per image without any associated confidence score or ranking mechanism. In practical deployment, operators would benefit from confidence-ranked localization outputs that distinguish high-certainty anomaly detections from marginal responses near the adaptive threshold, enabling prioritization of inspection effort.

Fourth, the evaluation is bounded to a single product type photographed under controlled frontal-view conditions with a uniform background. The HOG+LBP+SVM pipeline and the autoencoder localization module were both designed and optimized for this specific visual domain. Generalization to other casting geometries, multi-angle acquisition setups, variable illumination conditions, or different surface materials would require dataset-specific retraining and potentially hyperparameter adjustment for both stages.

Fifth, the computational cost of the Non-Local Means denoising step, which performs self-similar patch search over a 21×21 search window, is substantially higher than that of bilateral filtering alone. In high-throughput production line settings where per-image inference latency is constrained, this computational overhead may limit deployment viability without hardware acceleration or algorithmic approximation.

4.3 Future Research Directions

Building on the findings and limitations of this study, several productive directions for future research are identified.

The noise robustness evaluation framework could be extended to a continuous spectrum of noise intensities and multiple noise types, enabling a systematic characterization of the pipeline degradation and recovery profile across realistic industrial degradation profiles. An adaptive noise estimation module that measures the approximate noise level of each input image and dynamically selects denoising parameters accordingly could further improve cross-condition generalization without requiring separate model training for each noise configuration.

The spatial distortion robustness analysis could be extended to include rotation, affine

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transformation, perspective distortion, and motion blur, providing a more comprehensive mapping of the pipeline robustness envelope. Correspondingly, training augmentation strategies could incorporate these additional distortion types alongside zoom-and-crop, potentially enabling multi-axis robustness from a single unified augmentation scheme.

For Stage 2 localization, incorporating confidence scoring for individual bounding boxes would substantially improve practical utility. Per-region anomaly confidence scores could be derived from the mean or integrated reconstruction error within each detected region, providing a quantitative ranking of detected anomalies that operators could use to prioritize visual inspection of the most severely affected casting surface areas. Replacing the current bounding box formulation with instance segmentation would further improve spatial precision by producing pixel-level anomaly masks rather than rectangular approximations.

The integration of lightweight learned feature components with the hand-crafted pipeline represents a promising direction for improving domain generalization. A shallow pretrained convolutional backbone used as a fixed feature extractor, combined with the existing SVM classification head, could yield feature representations that are more invariant to both noise and spatial distortions than raw HOG and LBP descriptors while preserving the interpretability and data efficiency advantages of the current approach [15].

Deployment-oriented studies should investigate the end-to-end throughput and per-image latency of the proposed pipeline under production line speed constraints, identifying computational bottlenecks and evaluating real-time-compatible algorithmic alternatives for the NL-Means denoising step and the autoencoder inference stage. Quantization or knowledge distillation applied to the autoencoder could reduce inference cost while preserving localization accuracy [16].

Finally, extending the evaluation to additional casting datasets from different manufacturing contexts would provide broader evidence for the generalizability of the domain-bridging noise-hardening strategy and the unsupervised autoencoder-based localization approach beyond the single-product, single-viewpoint setting examined in this study.

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Appendix

- Source Code
- Dashboard
- Repository Github

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